

976. Largest Perimeter Triangle:

Given an integer array `nums`, return *the largest perimeter of a triangle with a non-zero area, formed from three of these lengths*. If it is impossible to form any triangle of a non-zero area, return `0`.

Example 1:

Input: `nums = [2,1,2]`

Output: 5

Explanation: You can form a triangle with three side lengths: 1, 2, and 2.

Example 2:

Input: `nums = [1,2,1,10]`

Output: 0

Explanation:

You cannot use the side lengths 1, 1, and 2 to form a triangle.

You cannot use the side lengths 1, 1, and 10 to form a triangle.

You cannot use the side lengths 1, 2, and 10 to form a triangle.

As we cannot use any three side lengths to form a triangle of non-zero area, we return 0.

Constraints:

- `3 <= nums.length <= 104`
- `1 <= nums[i] <= 106`

Solution: Time = $O(n \log n)$ and Space = $O(1)$

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class Solution:
    def largestPerimeter(self, nums: List[int]) -> int:
        nums.sort()
        for i in range(len(nums)-1, 1, -1):

            a = nums[i - 2]
            b = nums[i - 1]
            c = nums[i]
            if a + b > c:
                return a+b+c

        return 0
```